

Beyond 7/5 for Graphic TSP on Bipartite Graphs

Shiva Kintali*

September 12, 2026

Abstract

We give a deterministic polynomial-time $53/38$ -approximation for graphic TSP on every connected unweighted bipartite graph. Our algorithm computes a minimum binary even core F , covers its isolated vertices by removable private squares, separates intersecting cycles by vertex copies, and applies four tour constructions to the resulting integer pseudometric.

1 Introduction

Graphic TSP asks for a shortest closed walk visiting every vertex of a connected unweighted graph, equivalently a minimum-size connected spanning Eulerian multigraph. Let G be a connected unweighted bipartite graph. For the optimum and our tour lengths $\text{OPT}(G)$ and $\text{ALG}(G)$, we prove

$$\text{ALG}(G) \leq \frac{53}{38} \text{OPT}(G), \quad \frac{53}{38} = \frac{7}{5} - \frac{1}{190}.$$

Christofides and Serdyukov proved the classical $3/2$ guarantee for metric TSP [2, 13]; Karlin, Klein, and Oveis Gharan later obtained a randomized $(3/2 - \epsilon)$ approximation [6]. For graphic TSP, successive bounds below $3/2$ were obtained by Oveis Gharan, Saberi, and Singh [11], Mömke and Svensson [8], and Mucha [9]. Sebő and Vygen’s $7/5$ -approximation applies to all connected graph metrics and hence to bipartite graphs [12]; we improve it by $1/190$.

Stronger ratios were proved for special classes. Karp and Ravi obtained $9/7$ for cubic bipartite graphs and $9/7 + 1/[21(k - 2)]$ for k -regular bipartite graphs with $k \geq 4$ [7]; van Zuylen improved the cubic bipartite ratio to $5/4$ [14]. Related bounds for subquartic or cubic graphs appear in [10, 1, 4, 15].

Our proof has four steps.

- (i) One ordinary minimum-weight perfect matching finds a binary even core F minimizing $\Phi = |F| + 2i(F)$, with $n \leq \Phi \leq \text{OPT}(G)$ on cyclic inputs.
- (ii) Private squares cover the i isolated vertices, and vertex copies separate cycles meeting at a vertex; both operations admit exact projections back to the input graph.
- (iii) Four polynomial tour constructions P_1, P_2, Q, D are compared with one Hamiltonian benchmark. Matching gadgets encode their local debit functions, while compatible phase switches control connectivity.

*Email: shiva.kintali@gmail.com Homepage: shivakintali.github.io

- (iv) When the augmented decomposition has at least two cycles, averaging their bounds with weights $(8, 6, 1, 4)/19$ gives

$$\text{ALG}(G) \leq \frac{37}{38}\Phi(G) + \frac{8}{19}\text{OPT}(G) - \frac{7}{19}i - 2.$$

The one-cycle and exceptional square cases are solved optimally.

The augmented core has $N = |F| + 4i$ formal vertices and comparison cost $O = \text{OPT}(G) + 4i$. The common-comparator theorem gives $U \leq (37/38)N + (8/19)O - 2$; identifying copies and deleting the private squares saves $4i$ and yields (iv). Sections 2–5 develop the core, algorithms, and comparison, and Section 6 performs the transfer and analyzes the running time.

2 Problem and main result

Let $G = (V, E)$ be a finite connected simple unweighted undirected graph, $n = |V|$, and $m = |E|$. A *graphic TSP tour* is a multiplicity vector $x \in \mathbb{Z}_{\geq 0}^E$ for which the graph $(V, \{e \in E : x_e > 0\})$ is connected and

$$\deg_x(v) = \sum_{e \ni v} x_e$$

is positive and even at every vertex when $n \geq 2$. Its cost is $\sum_{e \in E} x_e$. Equivalently, it is a connected spanning Eulerian multigraph supported on G . For $n = 1$ we allow the empty tour, of cost zero. Write $\text{OPT}(G)$ for the minimum cost. For the algorithm stated below, $\text{ALG}(G)$ denotes the cost of its output.

For $n \geq 3$, the optimum equals that of symmetric metric TSP in the complete graph with shortest-path distances d_G . Indeed, shortcutting repeated vertices from an Euler circuit of a graphic tour gives a metric Hamilton cycle without increasing its length. Conversely, replacing every edge of a metric Hamilton cycle by a shortest path in G gives a connected spanning multigraph of the same cost: every Hamilton-cycle vertex has two path endpoints, and every internal occurrence on an expanded path adds degree two, so all degrees are positive and even. Since distinct vertices have distance at least one, every metric Hamilton cycle has cost at least n . Thus

$$\text{OPT}(G) \geq n \quad (n \geq 3). \tag{1}$$

Definition 2.1 (Binary even core). An edge set $F \subseteq E$ is *binary even* if every degree in the spanning subgraph (V, F) is even, with degree zero allowed. Define

$$\begin{aligned} f(F) &= |F|, & i(F) &= |\{v \in V : \deg_F(v) = 0\}|, \\ \Phi(F) &= f(F) + 2i(F), & \Phi(G) &= \min_{F \text{ binary even}} \Phi(F). \end{aligned}$$

The word *binary* distinguishes the 0–1 edge set F from a tour, which may use an edge more than once. Positive degrees in (V, F) need not equal two, so a binary even core need not be a two-factor.

Theorem 2.2 (Bipartite graphic TSP). *There is a deterministic polynomial-time algorithm which, for every connected unweighted bipartite graph G , constructs a graphic TSP tour with*

$$\text{ALG}(G) \leq \frac{53}{38}\text{OPT}(G), \quad \frac{53}{38} = 1.3947368421 \dots = \frac{7}{5} - \frac{1}{190}.$$

More precisely, suppose that G contains a cycle. Let F be the actual minimum- Φ core returned by the matching algorithm below, let $f = |F|$, $i = i(F)$, and $\Phi = f + 2i$, and let $\mathcal{C}$ be the edge-disjoint simple-cycle decomposition of F produced by Lemma 2.4. If $|\mathcal{C}| + i \geq 2$, then the output satisfies

$$\text{ALG}(G) \leq \frac{37}{38}\Phi + \frac{8}{19}\text{OPT}(G) - \frac{7}{19}i - 2 \leq \frac{53}{38}\text{OPT}(G) - 2. \quad (2)$$

Trees are handled exactly. If $|\mathcal{C}| + i = 1$, then $i = 0$ and F is a single spanning cycle, which is also returned optimally. When this cycle is a square, the additive -2 inequality need not hold; the algorithm is still optimal, so the multiplicative guarantee holds with ratio one.

The proof, completed in Subsection 6.3, is constructive and uses ordinary weighted perfect matching as a standard polynomial primitive [5].

2.1 Computing an even core and certifying its lower bound

Proposition 2.3 (Minimum even core). *One minimum-weight perfect-matching problem computes a minimum- Φ core; the auxiliary graph has $4m$ vertices, $O(m^2)$ edges, and weights in $\{-2, 0, 1\}$. If G contains a cycle, then*

$$n \leq \Phi(G) \leq \text{OPT}(G),$$

and every minimum core is nonempty. More generally, every binary even core F satisfies $|F| \geq n - i(F)$ and $\Phi(F) \geq n + i(F)$.

Proof. For each input vertex v of degree d_v , create a set P_v of d_v ports, one port $p_{v,e}$ for every incident edge e , and a set S_v of d_v slots. Join every vertex of P_v to every vertex of S_v by an edge of weight zero. Make S_v a zero-weight clique, except that, when $d_v \geq 2$, give one distinguished slot edge weight -2 . Finally, for every original edge $e = uv$, add the external edge $p_{u,e}p_{v,e}$ of weight one. There are no other edges.

Consider any perfect matching of this auxiliary graph, and let t_v be the number of ports in P_v matched by external edges. The other $d_v - t_v$ ports can be matched only to distinct slots, so precisely t_v slots remain to be matched within S_v . Hence t_v is even. Selecting an original edge exactly when its external edge is used therefore produces a binary even core F , with $t_v = \deg_F(v)$.

Conversely, fix a binary even core F . Use its external edges first. If $t_v = 0$, match all ports in P_v bijectively to S_v . If $t_v \geq 2$, leave the two distinguished slots free, match the $d_v - t_v$ unused ports to other slots, use the distinguished edge, and pair the remaining $t_v - 2$ free slots arbitrarily inside the clique. This is possible because t_v is even and the port-slot graph is complete bipartite. Thus every binary even core extends to a perfect matching.

For a fixed F , an inactive vertex ($t_v = 0$) cannot use its distinguished edge, because all of its slots are needed by ports. At an active vertex ($t_v \geq 2$), the construction above uses the unique negative edge and is locally optimal. Consequently the minimum perfect-matching weight among matchings encoding F is exactly

$$f(F) - 2(n - i(F)) = \Phi(F) - 2n,$$

because the external edges contribute $f(F)$ and precisely $n - i(F)$ vertices are active. Minimizing matching weight therefore minimizes $\Phi(F)$. The all-port-to-slot matching, which encodes the empty core, proves feasibility regardless of whether G has a perfect matching. The auxiliary graph has $2\sum_v d_v = 4m$ vertices and $O(\sum_v d_v^2 + m) = O(m^2)$ edges, since $\sum_v d_v = 2m$ and $\sum_v d_v^2 \leq 4m^2$. The one-vertex input is handled separately.

For the bounds, every nonisolated vertex of (V, F) has degree at least two. The degree-sum formula gives

$$2f(F) = \sum_{v \in V} \deg_F(v) \geq 2(n - i(F)).$$

Thus $f(F) \geq n - i(F)$ and $\Phi(F) = f(F) + 2i(F) \geq n + i(F) \geq n$. Taking a minimum over F proves $\Phi(G) \geq n$.

Choose an optimum graphic tour and let x_e be the multiplicity of edge e . We may assume $x_e \leq 2$ for every edge: if $x_e \geq 3$, deleting two copies leaves e in the support, preserves connectivity, subtracts two from the degrees of its endpoints, and strictly reduces the cost. Define

$$F_* := \{e \in E : x_e = 1\}, \quad D_* := \{e \in E : x_e = 2\}.$$

Reduction modulo two of the tour degrees shows that F_* is binary even, and

$$\text{OPT}(G) = |F_*| + 2|D_*|.$$

Let c_* be the number of components of (V, F_*) , including isolated vertices. Contracting these components in the connected tour support shows that the edges of D_* connect the resulting c_* vertices. Consequently $|D_*| \geq c_* - 1$.

It remains to show that F_* is nonempty. Choose a simple cycle C of length $\ell \geq 3$ in G and contract all its vertices to one vertex. The contracted graph is connected and has $n - \ell + 1$ vertices, so, after discarding loops, it has a spanning tree with $n - \ell$ edges. Expand the contraction and take one copy of every edge of C together with two copies of every tree edge. The doubled tree connects every vertex to C and contributes even degree at each endpoint; the cycle gives every vertex of C positive even degree. This is therefore a graphic tour of cost

$$\ell + 2(n - \ell) = 2n - \ell < 2n - 2.$$

If F_* were empty, every edge in the optimum support would instead have multiplicity two. A connected spanning support has at least $n - 1$ edges, so that tour would cost at least $2n - 2$, a contradiction. Hence $F_* \neq \emptyset$. The subgraph (V, F_*) therefore has at least one nonisolated component, and so $i(F_*) \leq c_* - 1 \leq |D_*|$. It follows that

$$\Phi(G) \leq |F_*| + 2i(F_*) \leq |F_*| + 2|D_*| = \text{OPT}(G).$$

Finally, the empty core has value $2n$, whereas the binary even core $E(C)$ has value $\ell + 2(n - \ell) = 2n - \ell < 2n$. Thus no minimum- Φ core is empty. $\square$

Lemma 2.4 (Even-cycle decomposition). *Every binary even core of a bipartite graph decomposes, in polynomial time, into edge-disjoint simple even cycles; the cycles may share vertices.*

Proof. Consider the subgraph formed by the edges not yet assigned to a cycle. Its degrees are even. If it has an edge, take a maximal trail in a nonempty component. The trail cannot end at a vertex different from its starting vertex: at such an endpoint the trail would use an odd number of incident edges, while maximality would say that it uses all of them, contradicting the even degree of that vertex. Hence the trail is closed. A shortest nonempty closed subtrail is a simple cycle.

Delete the edges of this cycle. Exactly two incident edges are removed at each of its vertices, so all remaining degrees are even. Iterating therefore partitions F into edge-disjoint simple cycles. At least one edge is removed per iteration, giving a polynomial-time algorithm. Every cycle in a bipartite graph is even, and simplicity of the input makes its length at least four. The cycles may share vertices because only their edges are deleted during the iteration. $\square$

3 Formal even-factor construction

The intermediate theorem is stated for exactly the distances needed by vertex copying, but not for an arbitrary weighted graphic instance.

Definition 3.1 (Admissible formal instance). An admissible instance consists of a finite set W , $N = |W|$, an integer pseudometric d , and a partition $\mathcal{B}$ of W into bags. Each bag is equipped with a designated cyclic order of even length at least four, and consecutive vertices in that order have distance one. Distinct vertices in the same bag have distance at least one. The graph on W joining distinct pairs with $d \leq 1$ is connected. A pseudometric is symmetric and nonnegative, satisfies the triangle inequality, and has $d(x, x) = 0$; distinct vertices may have distance zero.

The designated cyclic order and its unit edges form the designated cycle of a bag. Its *unit graph* contains every pair of distinct bag vertices at distance one, so additional unit chords are allowed. All local constructions use edges of this unit graph; a *crossing* joins two distinct bags. Formal multigraph edges always join distinct formal vertices. The output is a connected metric multigraph in which every vertex has positive even degree, and its cost is the sum of its edge-copy distances. Zero-distance edges between distinct bags are retained until the projection back to the original graph.

Theorem 3.2 (Even-factor comparison with signed crossings). *For an admissible instance with at least two bags, there is a deterministic polynomial algorithm with the following guarantee. Let H be any Hamilton cycle on W of cost $O \geq N$, and let h count designated squares having at least one vertex whose two neighbors in H lie outside that square. The algorithm does not need H , O , or h , but its output has cost*

$$U \leq \frac{37}{38}N + \frac{8}{19}O - \frac{2}{19}h - 2. \quad (3)$$

With one bag the designated spanning cycle is used directly and is optimal: every edge copy has cost at least one and positive even degrees require at least N edge copies. The additive formula is not asserted for a lone square.

The condition $O \geq N$ is not automatic for an arbitrary pseudometric; it will be verified in the bipartite reduction. Individual crossing excesses $d - 1$ may be negative even when $O \geq N$. None of them is discarded or replaced by zero.

3.1 Local covers and a global switching algorithm

For a bag $B = C_k$, a selected physical crossing matching exposes an even set $S \subseteq B$ of distinct vertices, called its ports. Put $|S| = 2j$. Each port is used by exactly one crossing edge. When $j > 0$, list the ports in cyclic order and let $\ell_0, \dots, \ell_{2j-1}$ be the positive lengths of the intervening cycle gaps. Define

$$a = a(S) = \min \left(\sum_{t \text{ even}} \ell_t, \sum_{t \text{ odd}} \ell_t \right), \quad \delta = k - 2a \in 2\mathbb{Z}_{\geq 0}.$$

The two alternating gap unions are the only subsets of designated cycle edges whose odd-degree set is S , so a is the minimum cycle S -join length. It is also the minimum perfect-matching cost on S in the cycle metric. For the lower bound, expand the edges of any such matching into cycle paths and reduce the resulting edge multiplicities modulo two; this produces a cycle S -join without increasing cost. For the upper bound, the gaps in the cheaper alternating union pair consecutive ports and have total length a .

Lemma 3.3 (Complementary rooted covers). *For $2j > 0$, there are two explicit spanning local multigraphs whose odd vertices are precisely S . Each has j connected pieces, each containing exactly two ports. Their costs are*

$$L_0 = k + a - 2j, \quad L_1 = 2k - a - 2j = L_0 + \delta.$$

Their induced port pairings P, Q have union one alternating cycle. For $S = \emptyset$ retain the whole cycle at cost k .

Proof. Choose one parity of gaps to retain once. In every gap of the other parity, delete an arbitrary edge and double all other edges. If the total length of the once-retained gaps is g , the cost is $g + 2(k - g - j) = 2k - g - 2j$. Taking $g = k - a$ or $g = a$ gives the two costs. The deleted edge in each complementary gap separates two doubled tails. Each once-retained gap, together with the attached doubled tails, is one piece connecting its two ports. Every nonport is covered and has even positive degree. The once-retained gap contributes one incident edge at every port, so its local degree is positive and odd. Adding its crossing edge therefore makes its total degree even and positive. The two pairings alternate consecutive ports around C_k , so their union is one alternating cycle, with the usual parallel-edge interpretation when $j = 1$. $\square$

Lemma 3.4 (Pairings and component decrease). *For perfect matchings P, Q, R on $2j$ points, let $c(P, R)$ count the alternating cycles in their multiset union. If $c(P, Q) = 1$, then*

$$c(P, R) + c(Q, R) \leq j + 1. \tag{4}$$

Consequently, if a switchable bag meets q current global components, switching its phase leaves its vertices in at most $j + 1 - q$ components and decreases the total component count by at least $2q - j - 1$.

Proof. A two-edge switch changes $c(\cdot, R)$ by at most one, while an alternating cycle of length $2t$ can be split into t common edges in $t - 1$ switches. Hence the switch distance between matchings A, B is $j - c(A, B)$. Its triangle inequality, applied to P, R, Q , gives (4).

Keep all crossing edges as formal metric edges, not expanded paths. Every local piece in a switchable bag is a once-retained path with doubled tails, hence has an Euler trail between its two ports. Replace each such two-terminal piece by a virtual edge, suppressing its interior and doubled excursions. In each current global component meeting this bag, take an Euler tour of the resulting connected even multigraph. Removing the bag's virtual edges from these tours produces outside trails and a pairing R of the ports with $c(P, R) = q$.

Insert the complementary pairing Q . By (4) it gives at most $c(Q, R) \leq j + 1 - q$ connected traversals. Intersections between outside trails or the restored local pieces can only merge these traversals, not create more actual components. Restoring doubled tails preserves the connectivity of each prescribed local piece. Components not previously meeting the bag are unchanged. Subtract the old count q . This argument uses the current components and Euler tours; it does not assume that component partitions monotonically coarsen under switches. $\square$

Call a bag *inactive* when $j = 0$ and *active* otherwise. An inactive bag retains its designated cycle. A two-port bag ($j = 1$) uses one fixed connected construction: either its cheaper cover from Lemma 3.3 or, when specified later, a certified endpoint walk. A multiport bag ($j \geq 2$) is *switchable* and uses the two phases from Lemma 3.3. Call it *free* if $\delta = 0$ and *paid* if $\delta > 0$.

Start every switchable bag in its cheaper phase, and then carry out the following deterministic procedure, choosing the first eligible bag in a fixed order.

1. Exhaust zero-cost phase changes that strictly decrease the actual number of connected components.
2. Repeatedly inspect *all* phase changes, in both directions. If a change decreases the actual component count by $d_c > 0$ and changes the local edge cost by at most d_c , execute it. Reinspect every bag, including free changes and negative-cost reverse changes. Stop when none is eligible.

Each change strictly decreases an integer component count, initially at most N , so fewer than N changes occur. A scan tests candidate multigraphs directly; no inferred compatibility of separate local partitions is used. Free rescans in the second phase matter: other switches can change a bag's incidences without coarsening the partition.

Let c_0, c_1 be the component counts after the first phase and at final termination. The first phase costs zero. By the stopping rule, the net increase in local cost during the second phase is at most $c_0 - c_1$. For each active bag B , write i_B, f_B for the number of components it meets at these two times, and let b_0 be the number of inactive bags. An inactive bag is one separate component throughout the switching procedure. Every other component meets at least two bags: every local piece in an active bag contains ports, and every selected crossing joins different bags. Therefore

$$\sum_{B \text{ active}} i_B \geq 2(c_0 - b_0), \quad \sum_{B \text{ active}} f_B \geq 2(c_1 - b_0).$$

The connected distance-at-most-one graph induces a connected graph on the final component quotient. Choose a spanning tree of that quotient and double its edges; this joins all final components, preserves even degrees, and adds cost at most $2(c_1 - 1)$. Let X be the selected crossing matching and set

$$E_X = \sum_{uv \in X} (d(u, v) - 1).$$

The matching X has $\sum_B j_B$ edges; its unit part is therefore charged as j_B per bag. Write $k_B = |B|$, and let L_B be the initial local cost. The completed tour has cost at most

$$\begin{aligned} U &\leq E_X + \sum_B (L_B + j_B) + (c_0 - c_1) + 2(c_1 - 1) \\ &\leq E_X - 2 + \sum_{B \text{ active}} \left(L_B + j_B + \frac{i_B + f_B}{2} \right) + \sum_{B \text{ inactive}} (k_B + 2). \end{aligned} \quad (5)$$

The first sum in the first line uses $j_B = 0$ for inactive bags. Adding the two displayed incidence inequalities gives

$$c_0 + c_1 \leq 2b_0 + \frac{1}{2} \sum_{B \text{ active}} (i_B + f_B),$$

which proves the second line. The calculation remains valid when E_X is negative because the crossings never change during the phase algorithm.

Call the bracketed quantity a bag's *debit*. A fixed connected two-port construction of local cost L_B has $j_B = i_B = f_B = 1$, so its debit is $L_B + 2$. For the ordinary cheaper two-port cover this equals $k + a$. An inactive bag's debit is $k + 2$.

Lemma 3.5 (Multiport debit bounds). *For a bag with $j \geq 2$, the debit in (5) satisfies*

$$D_B \leq k + a - j/2 + \frac{1}{2} \min\{j, \lfloor (j + \delta)/2 \rfloor\} \quad (\delta > 0), \quad (6)$$

$$D_B \leq 3k/2 - j + \lfloor (j + 1)/2 \rfloor \quad (\delta = 0). \quad (7)$$

Also $D_B \leq k + a$ whenever the cheaper local cost is $k + a - 2j$.

Proof. Each phase has exactly j local pieces, and every component meeting the bag contains at least one of them; hence $i_B, f_B \leq j$. For a free bag, if its incidence q exceeds $\lfloor (j+1)/2 \rfloor$, Lemma 3.4 gives a strictly component-decreasing free change. This is excluded both after phase one and at termination. Thus both incidences obey that bound, giving (7).

For a paid bag ending in its cheaper state, an incidence $q > \lfloor (j+\delta)/2 \rfloor$ gives $2q - j - 1 \geq \delta$, making its positive-cost change affordable. This contradicts termination. A paid bag ending in its expensive state has a negative-cost reverse change; every strictly positive component decrease would be affordable. Its incidence is therefore at most $\lfloor (j+1)/2 \rfloor$, which is no larger than the claimed paid cap because $\delta \geq 1$. Use $i_B \leq j$ and the cheaper initial cost in (5) to obtain (6). Finally, using only $i_B, f_B \leq j$ gives $D_B \leq k + a$. The cost telescoping already pays for bags ending in an expensive phase; their cost is not charged again. $\square$

3.2 Polynomial certificates for the two-port exceptions

Only two-port antipodal states require special treatment. Two vertices of C_k are *antipodal* when their designated-cycle distance is $k/2$. The antipodal pairs form a perfect matching on the vertices of an even cycle. An *endpoint walk* for u, v means a connected spanning multigraph in the bag's unit graph whose odd-degree set is exactly $\{u, v\}$; it has an Euler trail from u to v . Additional unit chords can make some antipodal endpoint walks cheaper. We recognize exactly the improvements needed below without invoking a general Hamilton-path oracle.

Definition 3.6 (GOOD and BAD pairs). In a square, a pair is GOOD if its endpoints admit a spanning Hamilton path in the bag's unit graph; this is tested by enumerating all $4!$ vertex orders. For even $k \geq 6$, every nonantipodal pair is GOOD. An antipodal pair u, v is declared GOOD by the certification procedure below exactly when it supplies an endpoint walk of cost at most $3k/2 - 3$. All other pairs are BAD. Let $\text{Bad}(S) = 1$ exactly when S consists of one BAD pair, and zero otherwise. Let $\text{Bad}_0(S) = 1$ exactly when S consists of one antipodal pair, whether certified or not.

The graph of BAD pairs is a submatching of the cycle antipodal matching. For a square, the designated cycle supplies Hamilton paths between adjacent vertices, so only its two diagonals can be BAD. For $k \geq 6$, a nonantipodal pair has $a \leq k/2 - 1$, and its cheaper two-port cover is an endpoint walk of cost

$$k + a - 2 \leq 3k/2 - 3.$$

Every GOOD antipodal pair is used with the explicit witness returned by the procedure. The procedure has the following completeness property, which is needed in the comparison.

Lemma 3.7 (Antipodal certification). *The GOOD classification and its witnesses can be computed in polynomial time. For $k \geq 6$, every antipodal pair admitting a connected spanning unit-edge endpoint walk of length at most $3k/2 - 3$ is classified GOOD. For $k = 4$ the stated Hamilton-path test is exact.*

Proof. The $k = 4$ test is the finite enumeration in the definition. In a minimum-cost connected spanning unit-edge T -join, no edge needs multiplicity greater than two: deleting two copies of an edge used at least three times preserves its support, parity, and connectivity while reducing the cost. For $k = 6$, enumerate multiplicities 0, 1, 2 on the at most $\binom{6}{2}$ unit edges and test connectivity, positive degrees, the prescribed odd set, and the cost bound. This is a fixed finite computation, and the preceding deletion argument proves that the enumeration is exhaustive.

Now let $k \geq 8$. Give designated cycle edges modified length one and every other edge of the bag's unit graph modified length two. Let ρ be the resulting shortest-path metric. For antipodal u, v , put $D_* = \rho(u, v)$. The two designated half-cycles show $D_* \leq k/2$, and D_* is an integer.

If $D_* \leq k/2 - 1$, take a shortest simple modified path J . If it uses a designated cycle edge e , form the original unit-edge multigraph $C_k + J$ and remove both copies of e . It still contains the spanning path $C_k - e$, so it is connected and spanning, with odd set $\{u, v\}$. Its unit cost is at most

$$k + |E(J)| - 2 \leq k + D_* - 2 \leq 3k/2 - 3.$$

If J consists only of q chords, then $2q = D_* \leq k/2 - 1$, and

$$q \leq \lfloor k/4 - 1/2 \rfloor \leq k/2 - 3 \quad (k \geq 8 \text{ even}).$$

The connected multigraph $C_k + J$ has cost $k + q \leq 3k/2 - 3$. Thus this branch always produces a witness.

It remains to handle $D_* = k/2$. Write $d_0 = k/2$ and $L_t = \{x \in B : \rho(u, x) = t\}$. A vertex in position t along either designated u - v half-cycle has ρ -distance exactly t from u : a shorter path to it followed by the remaining part of that half-cycle would give $\rho(u, v) < d_0$. Hence the distance layers are

$$|L_0| = |L_{d_0}| = 1, \quad |L_t| = 2 \quad (1 \leq t < d_0).$$

Every edge of the unit graph has modified length at most two. The triangle inequality for ρ therefore implies that its endpoints lie in layers whose indices differ by at most two. The sets $L_t \cup L_{t+1} \cup L_{t+2}$ for $0 \leq t \leq d_0$, with nonexistent end layers omitted, form a path decomposition: every vertex occurs in a consecutive set of these bags, and every unit edge occurs in at least one bag. Each bag has at most six vertices, hence the width is at most five. Compute an exact minimum connected spanning $\{u, v\}$ -join on this decomposition by Lemma 3.8. Classify the pair GOOD precisely if its cost is at most $3k/2 - 3$.

The low-distance branch succeeds unconditionally; the other branch computes the exact optimum. Together they prove the asserted completeness. $\square$

Lemma 3.8 (Explicit bounded-width endpoint-join dynamic program). *Given a graph on at least two vertices with a path decomposition of width at most five, an even prescribed odd set T , and unit edge costs, one can construct an exact minimum connected spanning T -join in polynomial time, or certify that none exists.*

Proof. No optimum uses an edge more than twice: from multiplicity at least three, deleting two copies preserves positive support, parity, and connectivity and strictly decreases the unit cost. Thus it suffices to consider multiplicities 0, 1, 2.

Refine the decomposition to introduce-vertex, introduce-edge, and forget-vertex operations, with at most six boundary vertices, and introduce each edge while both endpoints are present. A state records, for every boundary vertex, its degree parity and whether its degree is positive, together with the partition induced by selected edges and one final-closure bit. On introducing an edge, try multiplicities 0, 1, 2 and update these data and the cost. A forgotten vertex must have parity $\mathbf{1}[v \in T]$ and positive degree; reject a transition that removes the last boundary representative of a selected component. The sole exception is the terminal transition, after every vertex and edge has been processed, when the unique selected component may close; the closure bit records this event and no later transition is permitted.

The usual induction on the operations shows that these states encode exactly the partial connected spanning T -joins: a component that has left the boundary can acquire no later edge, while every feasible join induces a nonrejected state sequence. There are at most $2^6 2^6 B_6$ states, up to the closure bit, and constant branching per operation. Minimum-cost predecessor pointers therefore give the optimum and a witness in polynomial time. $\square$

4 Matching algorithms for the debit functions

4.1 Three long-cycle debit functions and the square debit

Fix a bag $B = C_k$, an even port set S with $|S| = 2j$, and the parameters $a = a(S)$ and $\delta = k - 2a$ from Subsection 3.1. In the linear and precise modes, a GOOD two-port state uses its certified connected endpoint walk; every other two-port state uses the cheaper cover of Lemma 3.3. In the affine mode, the same rule is used except that every antipodal pair is left unrescued. A state with at least four ports uses the two switchable phases of Lemma 3.3, and an empty state retains C_k . Every fixed two-port construction is connected, so its incidence is one at both times in the ledger (5).

Lemma 4.1 (Represented local bounds). *The switching construction satisfies each applicable fixed debit bound below:*

$$D_{\text{lin}}(S) \leq \frac{3}{2}k - \frac{j}{4} - \frac{1}{4}\mathbf{1}[S \neq \emptyset] + \frac{1}{2}\text{Bad}(S) \quad (k \geq 6), \quad (8)$$

$$D_{\text{prec}}(S) \leq \frac{3}{2}k - 1 + \text{Bad}(S) \quad (k \geq 6), \quad (9)$$

$$D_{\text{aff}}(S) \leq \frac{5}{4}k + \frac{1}{2}a(S) - \frac{j}{4} + \frac{1}{4}\text{Bad}_0(S) \quad (k \geq 8). \quad (10)$$

In the affine formula take $a(\emptyset) = 0$. For a square, all three algorithms use

$$D_4(S) \leq 6 - A_4(S), \quad A_4(S) = \begin{cases} 1, & |S| = 4 \text{ or } S \text{ is a GOOD pair,} \\ 0, & \text{otherwise.} \end{cases} \quad (11)$$

For a hexagon the third algorithm may instead use the fixed bound nine for every state.

Proof. We check all possible cardinalities of S . For the linear bound, first suppose $j \geq 2$ and $\delta > 0$. Substituting $a = (k - \delta)/2$ in (6) and using $\min\{j, \lfloor (j + \delta)/2 \rfloor\} \leq (j + \delta)/2$ gives

$$D_B \leq 3k/2 - j/4 - \delta/4 \leq 3k/2 - j/4 - 1/4.$$

If $j \geq 2$ and $\delta = 0$, the saving in (7) relative to $3k/2 - j/4$ is $3j/4 - \lfloor (j + 1)/2 \rfloor \geq 1/4$ for every $j \geq 2$: for $j = 2t$ it is $t/2$, and for $j = 2t + 1 \geq 3$ it is $t/2 - 1/4$. For $j = 1$, a nonantipodal pair has $a \leq k/2 - 1$ and hence debit $k + a \leq 3k/2 - 1$. A certified antipodal endpoint walk has local cost at most $3k/2 - 3$ and therefore debit at most $3k/2 - 1$. Both estimates are stronger than the GOOD right side $3k/2 - 1/2$. A BAD antipodal pair has debit $k + a = 3k/2$, exactly the right side after the $\text{Bad}(S)/2$ term is included. Finally, $S = \emptyset$ has debit $k + 2 \leq 3k/2$ because $k \geq 6$. This proves (8).

For the precise bound, the empty debit satisfies $k + 2 \leq 3k/2 - 1$. The preceding two-port estimates give debit at most $3k/2 - 1$ for a GOOD pair and exactly $3k/2$ for a BAD pair. If $j = 2$ and $\delta > 0$, then δ is a positive even integer, so $k + a = 3k/2 - \delta/2 \leq 3k/2 - 1$; the coarse conclusion of Lemma 3.5 applies. If $j = 2$ and $\delta = 0$, equation (7) gives $3k/2 - 1$. If $j \geq 3$, then $\text{Bad}(S) = 0$ and (8) is at most $3k/2 - 1$. These cases prove (9).

For the affine bound with $j \geq 2$ and $\delta > 0$, (6) with the same relaxed minimum gives

$$D_B \leq 3k/2 - \delta/4 - j/4 = 5k/4 + a/2 - j/4.$$

For $\delta = 0$, (7) gives this bound since $\lfloor (j + 1)/2 \rfloor \leq 3j/4$. For a nonantipodal two-port set the right side minus its debit $k + a$ is $\delta/4 - 1/4 \geq 0$, because δ is positive and even. For an antipodal pair, $a = k/2$ and $\text{Bad}_0(S) = 1$, so the right side and the debit $k + a$ both equal $3k/2$. The empty case is $k + 2 \leq 5k/4$ for $k \geq 8$. This proves (10).

For a square, the only even port-set sizes are zero, two, and four. An empty square has debit six. A BAD pair has the ordinary debit $k + a = 6$, whereas a GOOD pair has a three-edge spanning path and hence debit $3 + 2 = 5$. With four ports, $j = 2$, $a = 2$, and $\delta = 0$, so (7) gives five. This proves (11). Finally, (9) with $k = 6$ bounds every hexagon debit by nine, including the empty and two-port states already checked. $\square$

The bounds are upper bounds on *actual completed output costs*. They are not independent connected-join objectives being minimized inside each bag after the ports have been chosen. The next construction realizes one fixed bound per bag in one ordinary matching problem.

4.2 Common physical/shadow architecture

We now encode one of the preceding debit bounds for every bag in a single ordinary minimum-weight perfect-matching instance. For each formal vertex v , create a physical vertex v and a shadow vertex $\bar{v}$, joined by a zero-weight *spoke*. Join physical vertices u, v in distinct bags by a *crossing edge* of weight $d(u, v) - 1$; there are no other edges incident with a physical vertex. Shadows have only their spoke and within-bag gadget edges. Every routing vertex introduced below is private to one bag.

Let M be a perfect matching of this graph. Each physical vertex uses either its spoke or one crossing edge, so the crossing edges of M form a matching X on formal vertices. In a bag B , let

$$S_B = \{v \in B : v \text{ is incident with an edge of } X\}.$$

Exactly the shadows of S_B remain to be matched inside the local gadget. Every gadget has an even number of auxiliary vertices; parity of the local perfect matching therefore forces $|S_B|$ to be even. Conversely, the lemmas below show that the local gadget can be completed for every even S_B .

The minimum weight of such a local completion is the gadget's *signature* $\sigma_B(S_B)$. Its constant baseline β_B is an accounting term and is not represented by an edge. For fixed crossings, the local gadgets are disjoint. Hence an optimal global matching uses a minimum local completion in every bag; otherwise replacing one completion would improve it. Consequently every optimum M^* satisfies the exact identity

$$w(M^*) = E_X + \sum_B \sigma_B(S_B). \quad (12)$$

Lemma 4.2 (Matching avoiding forbidden pairs). *Let the forbidden pairs of a finite set form a matching. Every even subset of size at least four has a perfect matching in the complete graph that contains no forbidden pair.*

Proof. Start with any perfect matching of the subset. If it contains a forbidden edge ab , choose another matching edge cd , which exists because the subset has size at least four. Replace ab, cd by ac, bd . Neither new edge is forbidden: in the forbidden-pair matching, a and b have no forbidden neighbors other than each other. Thus the number of forbidden edges strictly decreases. Repetition terminates with the desired matching. $\square$

Lemma 4.3 (Exact local signatures). *The following baselines and signatures are realized by ordinary matching gadgets with $O(k)$ vertices and $O(k^2)$ edges:*

mode	β_B	$\sigma_B(S)$
linear	$3k/2$	$-j/4 - \mathbf{1}[S \neq \emptyset]/4 + \text{Bad}(S)/2$
precise	$3k/2 - 1$	$\text{Bad}(S)$
affine	$5k/4$	$a(S)/2 - j/4 + \text{Bad}_0(S)/4$
hexagon	9	0
square	6	$-A_4(S)$.

The linear and precise rows apply for $k \geq 6$, the affine row for $k \geq 8$, the hexagon row for $k = 6$, and the square row for $k = 4$. Every gadget is feasible for every even S .

Proof. For the linear mode, partition the shadows into the two-element BAD classes and singleton classes. Join different classes at weight $-1/4$ and each BAD pair at weight zero. Add tokens s, t with edge st and, for every class C , routing vertices x_C, y_C with edge $x_C y_C$; these edges have weight zero. Join shadows of C to x_C at weight $-1/4$ and y_C to both tokens at weight zero. A perfect matching is either inactive, with all routing pairs and st used, or consumes exactly two shadows from distinct classes at cost $-1/2$. Lemma 4.2 then gives, for $|S| = 2j \geq 4$, one active pair and $j - 1$ ordinary pairs; the empty, GOOD-pair, BAD-pair, and multiport costs are respectively $0, -1/2, 0, -j/4 - 1/4$. At most one activation is possible and every remaining shadow edge has weight at least $-1/4$, so these costs are optimal and give the displayed signature.

For the precise mode, a shadow clique has weight one on BAD pairs and zero elsewhere. A two-shadow completion has cost $\text{Bad}(S)$, while Lemma 4.2 gives a zero-cost completion when $|S| \geq 4$. For the affine mode give uv weight

$$\frac{1}{2}d_{C_k}(u, v) - \frac{1}{4} + \frac{1}{4}\mathbf{1}[u, v \text{ antipodal}].$$

A minimum cycle-metric matching has length $a(S)$. If $|S| \geq 4$, it contains no antipodal edge: such an edge already has length $k/2 \geq a(S)$ and every other edge has positive length. Thus its cost is $a(S)/2 - j/4$; for one pair the unique edge adds precisely $\text{Bad}_0(S)/4$. Conversely, every completion has cycle-metric length at least $a(S)$ and nonnegative surcharge, proving exactness. A zero-weight clique realizes the hexagon row.

For a square, use the linear routing construction with a zero-weight shadow clique and shadow-to-routing weight $-1/2$. It activates once, at cost -1 , exactly for a GOOD pair or all four shadows. Hence its signature is $-A_4(S)$; the two tokens preclude a second activation. These constructions also prove feasibility and the stated sizes. $\square$

Proposition 4.4 (Four complete algorithms). *The following are four deterministic polynomial tour algorithms on an admissible instance:*

1. P_1 : square mode on squares and linear mode on every other bag;
2. P_2 : square mode on squares and precise mode on every other bag;
3. Q : square mode on squares, fixed constant-nine mode on hexagons, and affine mode on every bag of length at least eight;
4. D : retain all designated cycles and double a tree of their quotient in the distance-at-most-one graph.

Each of the first three uses one common perfect-matching optimization over all bags and crossings. Its output cost is at most its total baseline plus its optimum matching weight minus two.

Proof. Fix one of P_1, P_2, Q , and write β_B and σ_B for the baseline and signature selected for bag B . The all-spoke choice extends to a perfect matching: in the token gadgets match the two tokens together and every routing pair internally, while the other gadgets have no exposed shadows. Thus the global matching instance is feasible. Let M^* be a minimum-weight perfect matching, X its physical crossings, and S_B its even exposed set in bag B . Choose the local covers specified above, using the stored certificate whenever a GOOD two-port state requires one, and run the switching algorithm. The local debit inequality for every bag has the uniform form

$$D_B \leq \beta_B + \sigma_B(S_B).$$

This is (8), (9), (10), or (11); for the special hexagon mode it is the constant bound nine. Combining the global ledger with (12) gives

$$U \leq E_X - 2 + \sum_B D_B \leq -2 + \sum_B \beta_B + E_X + \sum_B \sigma_B(S_B) = -2 + \sum_B \beta_B + w(M^*),$$

which is the asserted objective bound.

For completeness, the decoded object is a valid tour. Each nonport has positive even local degree, and each port has positive odd local degree plus one incident crossing, hence positive even total degree. Switching preserves these properties. Doubling a spanning tree of the final component quotient makes the multigraph connected and preserves even degrees, so it has an Euler tour. All choices can be made deterministically by fixed tie-breaking. After multiplying weights by four, the matching instances have integral weights of polynomial bit length; the gadgets, certificates, switching process, and quotient tree are all polynomial-size and polynomial-time constructions, as detailed in Subsection 6.4.

For D , retain every designated cycle. Contracting the bags in the connected distance-at-most-one graph gives a connected quotient; double a spanning tree of that quotient. The result is again connected with every vertex of positive even degree and is constructed deterministically in polynomial time. $\square$

5 Common-comparator analysis

Fix the same Hamilton cycle H from Theorem 3.2 for the analysis of every algorithm. Since there are at least two bags, the edges of H internal to any fixed bag form a proper spanning linear forest: nontrivial vertex-disjoint paths together with isolated vertices. Call these paths and isolated vertices the *visits* of H to the bag. Let s_B , abbreviated to s when the bag is fixed, count its nontrivial visits, let m_B count its singleton visits, and put $t_B = s_B + m_B$. There are $2s$ distinct nontrivial visit endpoints. A singleton has two crossing edges in H , an endpoint has one, and a strict visit interior has none. Consequently the bag has $2t_B$ incidences with original H -crossing edges.

We need a legal matching comparison preserving a single global cycle, not independently chosen pairings whose union could have an odd cycle. The following lemma provides that compatibility.

5.1 The elementary preserving exchange

For $s > 0$, remove the interiors of the bag's nontrivial visits from the current main cycle and suppress every resulting outside path to one edge. The outside paths induce a perfect matching R_B on the $2s$ visit endpoints. A *shadow path cover* consists of exactly s vertex-disjoint paths on the $2s$ endpoint shadows and the m_B singleton shadows, with the prescribed endpoint shadows as its $2s$ path endpoints. It is *compatible* if, after suppressing its path interiors, its endpoint pairing together with R_B is one alternating cycle. Such a cover always exists: use the pairing induced by the original nontrivial visits, whose union with R_B reconstructs the current main cycle, and insert each singleton shadow into an arbitrary cover edge.

Lemma 5.1 (Compatible covers). *Every compatible shadow path cover can be changed to contain at most one BAD edge. If every internal H -edge in the bag has distance one, it can be chosen with no BAD edge; when all visits are singletons, use instead a separate designated even shadow cycle.*

Proof. If a compatible cover contains a BAD edge ab and a disjoint edge cd , remove both. Of the two alternative reconnections, exactly one preserves the single cycle with R_B ; it also preserves the

path endpoints (reversing the intervening subpath when both edges lie on one path). Neither new edge is BAD because the BAD pairs form a matching. Repeating this exchange leaves at most one BAD edge.

Assume now that all internal H -edges are unit. For $s \geq 2$, a BAD edge has a disjoint edge on another nontrivial shadow path, so the exchange removes every BAD edge. It remains to treat $s = 1$. Let the unique nontrivial visit have endpoints a, b and let $m = m_B$.

If $m = 0$, the visit is a spanning path of length $k - 1$. For $k \geq 6$, $k - 1 \leq 3k/2 - 3$, so if a, b are antipodal Lemma 3.7 certifies them. The square case follows from the exact Hamilton-path definition.

If $m = 1$, call the singleton u and use the shadow path a, u, b . Only an antipodal pair can be BAD. Suppose u, a are antipodal. Appending a shortest $b-u$ cycle path to the length- $(k - 2)$ visit gives a spanning $a-u$ endpoint walk of length at most $(k - 2) + (k/2 - 1) = 3k/2 - 3$; the other endpoint edge is symmetric. For $k = 4$ the resulting three-edge tree is a Hamilton path, so the exact square test accepts.

If $m = 2$, the two candidate paths are a, u, v, b and a, v, u, b . Every BAD endpoint edge has a disjoint edge in the same path, so repeated compatible exchanges remove all BAD endpoint edges. Any remaining BAD edge is the middle pair u, v ; if it is antipodal, the two attachment sums

$$d_{C_k}(u, a) + d_{C_k}(b, v), \quad d_{C_k}(u, b) + d_{C_k}(a, v)$$

sum to k . Attaching the cheaper pair to the length- $(k - 3)$ visit gives a spanning $u-v$ endpoint walk of length at most $3k/2 - 3$, including the square case just discussed.

If $m \geq 3$, every edge of the shadow path has a disjoint path edge, so the exchange removes every BAD edge. Finally, if $s = 0$, put the shadows on the designated even cycle; all its edges are nonantipodal and hence GOOD. $\square$

5.2 Building the actual half-marginal support

Lemma 5.2 (Common support and its matching distribution). *There is a distribution over feasible physical crossing sets and completions of the auxiliary perfect matchings such that:*

1. *each crossing edge of H is selected with probability $1/2$;*
2. *in each bag, $\mathbb{E}j_B = t_B/2$;*
3. *every bag is active with probability at least $1/2$, and every bag containing a singleton is active with probability one;*
4. *a bag with unit internal H -edges has probability zero of a BAD two-port state;*
5. *any other bag has probability at most $1/2$ of a BAD two-port state.*

The distribution is only a comparator for optimum ordinary matchings; the algorithm never needs to find H .

Proof. We construct a spanning support graph on the physical and shadow vertices, maintaining one distinguished main cycle. Initially this is H on the physical vertices. Process the bags sequentially. If $s > 0$, replace the bag's nontrivial physical visit paths by a compatible shadow path cover. At each of its $2s$ prescribed endpoints, use the spoke from the physical endpoint to its shadow. Leave each singleton physical vertex in its original position between its two crossing edges, while using its distinct shadow in the replacement cover. Remove every strict visit-interior physical vertex from

the main cycle and fix its spoke to its shadow; these fixed spokes are vertex-disjoint from the main cycle. Use Lemma 5.1 to leave no BAD shadow edge when all original internal edges are unit, and at most one otherwise. Compatibility makes the updated distinguished support a single cycle after every step.

If $s = 0$, leave all k physical vertices of the bag in their original positions on the main cycle and put their shadows on a separate copy of the designated even cycle. After all bags are processed, the support is the disjoint union of the main cycle, these separate shadow cycles, and the fixed spokes. It covers every physical vertex, every shadow, and no vertex twice. If $\mathcal{A}$ is the set of all-singleton bags, the number of vertices on the main cycle is

$$2 \sum_{B \notin \mathcal{A}} (2s_B + m_B) + \sum_{B \in \mathcal{A}} k_B.$$

For $B \notin \mathcal{A}$, the factor two counts separately the physical and shadow copies of its $2s_B$ endpoints and m_B singletons. The display is even because every k_B is even, and every separate shadow cycle is even as well. Independently on each support cycle, choose either of its two alternating perfect matchings with probability $1/2$, and include all fixed spokes. This is a perfect matching of all physical and shadow vertices before the routing vertices of the actual gadgets are inserted.

Every original crossing belongs to the main cycle, so its marginal is $1/2$. A nontrivial visit endpoint is incident on the main cycle with one crossing and one spoke, and therefore is exposed with probability $1/2$. A singleton physical vertex is incident there with its two original crossings and hence is exposed in either alternating phase. Thus

$$\mathbb{E}|S_B| = (2s_B)/2 + m_B = t_B, \quad \mathbb{E}j_B = t_B/2.$$

Every bag has at least one original crossing edge, whose selection alone implies activity, so its activation probability is at least $1/2$. A singleton makes activity certain. The support matching also pairs the exposed shadows inside each bag, directly confirming that every S_B is even.

All shadow edges in the support of a unit-forest bag are GOOD. In any other bag the support has at most one BAD shadow edge, and that edge has selection probability $1/2$. If S_B consists of a BAD pair, the support matching must match its two exposed shadows to each other, so it must have selected that unique BAD edge. Claims four and five follow.

Finally retain the selected physical crossings and replace, independently in each bag, the conceptual shadow matching by a canonical minimum completion of the fixed gadget for the algorithm under comparison. Every exposed set is even, so Lemma 4.3 gives such a completion. The substitution changes neither the crossings nor their marginals and realizes the exact signature of each exposed set. Thus it produces a distribution of feasible global matchings for each of the fixed objectives of P_1 and P_2 , without presuming any additional routing choice. $\square$

5.3 The two half-marginal comparison inequalities

Let r be the number of square bags, c the number of other bags, and L the total number of vertices in the latter; then $N = 4r + L$. Let $T = \sum_{B: k_B \geq 6} t_B$ be the total number of nonsquare visits. Let M_4, M_L be the numbers of singleton vertices in square and nonsquare bags respectively. Let h count squares with at least one singleton, and let α count nonsquare bags whose visits are all singletons. The symbol α is deliberately distinct from the local cycle-join length $a(S)$.

Put $e = O - N \geq 0$. Separate the original H excess into internal and crossing contributions:

$$e_I = \sum_{\substack{uv \in H \\ u, v \text{ same bag}}} (d(u, v) - 1) \geq 0, \quad e_X = \sum_{\substack{uv \in H \\ u, v \text{ different bags}}} (d(u, v) - 1), \quad e = e_I + e_X.$$

Here e_X may be negative. A *contaminated bag* contains an original nonunit internal H -edge. Since same-bag distances are positive integers, if b_4, b_L count contaminated square and nonsquare bags, then each such bag contains at least one unit of internal excess and

$$b_4 + b_L \leq e_I. \quad (13)$$

Lemma 5.3 (The two fixed P comparisons). *The actual outputs of P_1 and P_2 satisfy*

$$P_1 \leq \frac{11}{2}r + \frac{3}{2}L - \frac{T}{8} - \frac{c}{8} - \frac{\alpha}{8} - \frac{h}{2} + \frac{e}{2} - 2, \quad (14)$$

$$P_2 \leq \frac{11}{2}r + \frac{3}{2}L - c - \frac{h}{2} + \frac{e}{2} - 2. \quad (15)$$

Proof. Use Lemma 5.2 as a feasible comparison for each matching objective. Since $A_4(S) = \mathbf{1}[S \neq \emptyset] - \text{Bad}(S)$, square activation is at least $r/2 + h/2$ and its expected BAD count is at most $b_4/2$. The expected contributions, before the common -2 , are therefore

	P_1	P_2
squares	$\frac{11}{2}r - \frac{h}{2} + \frac{b_4}{2}$	$\frac{11}{2}r - \frac{h}{2} + \frac{b_4}{2}$
nonsquares	$\frac{3}{2}L - \frac{T}{8} - \frac{c}{8} - \frac{\alpha}{8} + \frac{b_L}{4}$	$\frac{3}{2}L - c + \frac{b_L}{2}$
crossing excess	$e_X/2$	$e_X/2$

Here $\mathbb{E}j_B = t_B/2$; activity totals at least $c/2 + \alpha/2$; and only a contaminated bag can realize a BAD state, with probability at most $1/2$. Finally, $b_4/2 + b_L/4 \leq e_I/2$ for P_1 and $(b_4 + b_L)/2 \leq e_I/2$ for P_2 by (13). Adding $e_X/2$, using $e = e_I + e_X$, and invoking Proposition 4.4 gives (14)–(15). $\square$

5.4 The shortcut comparison, including zero distances

The third algorithm has one important state-independent property: its represented nonsquare contribution is at most $3k/2$ for every exposed set. For hexagons this is the chosen baseline nine. For $k \geq 8$ and nonempty S , the exact affine contribution satisfies

$$\frac{5}{4}k + \frac{1}{2}a(S) - \frac{j}{4} + \frac{1}{4}\text{Bad}_0(S) \leq \frac{3}{2}k + \frac{-j + \text{Bad}_0(S)}{4} \leq \frac{3}{2}k,$$

using $a(S) \leq k/2$, $j \geq 1$, and $\text{Bad}_0(S) \leq 1$. The empty state has contribution $5k/4 \leq 3k/2$. This is a bound on the *same fixed Q objective*, not a new per-state choice between different algorithms.

Lemma 5.4 (Signed singleton-deletion comparison). *The actual Q output satisfies*

$$Q \leq 5r + \frac{3}{2}L + h + M_4 + M_L + e - 2. \quad (16)$$

Proof. Delete every vertex that was a singleton in its original H -bag, and shortcut each maximal deleted segment by the direct metric edge between its surviving endpoints. Call the resulting cyclic order H' . We first treat the case in which at least two bags contain surviving vertices.

Each surviving vertex retains an original internal H neighbor, since both endpoints of every original nontrivial visit survive. It is therefore incident to at most one interbag edge of H' . Select all such interbag edges. They form a physical matching. Moreover, every bag has an even number of their endpoints: along the cycle H' , every entrance into a bag is paired with an exit, while an internal edge contributes no boundary endpoint. Hence these edges define a feasible even exposed set for every bag in the Q matching instance. Each surviving bag contains a nontrivial visit and therefore

at least two vertices. Thus at least four vertices survive, so no one- or two-vertex cyclic-order convention is needed.

Every nonsquare bag contributes at most $3k/2$. Consider a square with no original singleton and no original nonunit internal edge. Its original internal forest is either a unit spanning path P_4 or two disjoint unit paths P_2 . In the P_4 case, the complementary arc of H' meets another surviving bag, so it cannot have been replaced by an internal shortcut of the square. The two exposed vertices are therefore the endpoints of a unit spanning path and form a GOOD pair. In the $P_2 \cup P_2$ case, all four endpoints remain exposed unless a deleted segment creates an internal shortcut joining the two paths. At most one such shortcut can occur, because two would make those four vertices an entire cycle component of H' . If the shortcut is unit, the two original P_2 's and this edge form a unit spanning path, whose two remaining endpoints are GOOD.

Thus every square has represented contribution at most six, and it contributes at most five unless it has an original singleton, an original nonunit internal edge, or a new internal *nonunit* shortcut. Let q_4^+ count shortcuts of the last type. Counting edges can only overcharge a square that receives more than one. The total square contribution is therefore at most

$$5r + h + b_4 + q_4^+.$$

In particular, a newly internal unit shortcut receives no charge; this is essential because deleted segments can contain negative crossing excess.

We now prove the signed accounting inequality

$$b_4 + q_4^+ + E_{\text{cross}}(H') \leq M_4 + M_L + e. \quad (17)$$

Let $\mathcal{J}$ be the set of maximal deleted segments. For $J \in \mathcal{J}$, let $z_J \geq 1$ be its number of deleted vertices, s_J the total metric length of its $z_J + 1$ original edges, and d_J the length of its shortcut. Its deleted-vertex count plus its original edge excess is exactly

$$z_J + (s_J - (z_J + 1)) = s_J - 1,$$

which may be negative. The triangle inequality gives $d_J \leq s_J$. The charge assigned to this segment is bounded by $s_J - 1$ in each of the following exhaustive cases:

1. If the shortcut crosses bags, its new excess is $d_J - 1 \leq s_J - 1$, including $d_J = 0$.
2. If it is charged to q_4^+ , then $d_J \geq 2$, so its charge one is at most $s_J - 1$.
3. If it is an ignored internal shortcut, its distinct endpoints lie in the same bag, hence $d_J \geq 1$. Its zero charge is at most $s_J - 1$.

In the second case $d_J \geq 2$, which was used to infer $1 \leq s_J - 1$. The deleted segments and their deleted vertices are pairwise disjoint. Every original edge in such a segment is a crossing: an H -singleton has no internal incident edge, and two adjacent deleted vertices therefore cannot lie in the same bag. Each original crossing not in a deleted segment remains unchanged in H' and pays exactly its own signed excess. Hence the preceding three cases, summed over $\mathcal{J}$, account for $q_4^+ + E_{\text{cross}}(H')$ using precisely $M_4 + M_L + e_X$. Finally, the nonnegative internal excess is disjoint from those crossing charges and, by (13), pays one for each of the b_4 contaminated squares. This proves (17) without discarding any negative crossing term. Combine the square bound, the nonsquare bound, and (17); then use the matching optimum and subtract two as in Proposition 4.4. This proves (16) in this case.

If at most one bag survives, compare the *same* Q objective with its all-spoke matching. Its total baseline is at most $6r + 3L/2$, so, after the global -2 , inequality (16) follows once $r \leq h + M_4 + M_L + e$.

We prove the stronger inequality without e . If every square has a singleton, then $h = r$. Otherwise a square with no singleton survives in its entirety. It is the unique surviving bag, and every other bag is entirely deleted. Thus every other square is counted by h , so $h = r - 1$; because the formal instance originally had at least two bags, some other bag of size at least four was deleted, and $M_4 + M_L \geq 4$. The case $r = 0$ is immediate. Finally $e = O - N \geq 0$. This fallback uses only that global inequality and never assumes that an individual crossing excess is nonnegative. $\square$

Lemma 5.5 (Coupled singleton counts). *The comparator parameters satisfy*

$$M_4 \leq 4h, \quad M_L \leq T - c + \alpha.$$

Proof. A square contributes no singleton unless it is one of the h counted squares, and each such square has at most four vertices; hence $M_4 \leq 4h$. In a nonsquare bag with a nontrivial visit, $s_B \geq 1$ and therefore $m_B = t_B - s_B \leq t_B - 1$. In an all-singleton nonsquare bag, $m_B = t_B$. Summing over the $c - \alpha$ bags of the first kind and the α bags of the second gives

$$M_L \leq \sum_B t_B - (c - \alpha) = T - c + \alpha.$$

$\square$

Corollary 5.6 (The fixed Q and direct comparisons). *The two actual outputs obey*

$$Q \leq 5r + \frac{3}{2}L + 5h + T - c + \alpha + e - 2, \quad (18)$$

$$D \leq 6r + L + 2c - 2. \quad (19)$$

Proof. Substitute Lemma 5.5 into Lemma 5.4. For D , the designated cycles cost $N = 4r + L$. Contracting the $r + c$ bags in the connected distance-at-most-one graph gives a connected quotient, so it has a spanning tree with $r + c - 1$ edges, each of cost at most one. Doubling that tree costs at most $2(r + c - 1)$ and yields (19). $\square$

5.5 Exact cancellation and proof of the formal theorem

Proof of Theorem 3.2. Run P_1, P_2, Q, D and return their shortest actual output. Its cost is at most every convex combination of their four costs, in particular the one with weights

$$(w_1, w_2, w_Q, w_D) = \frac{1}{19}(8, 6, 1, 4).$$

The four bounds (14), (15), (18), and (19) use the same fixed Hamilton cycle H . Their coefficients, before division by 19, sum as follows:

parameter	r	L	T	c	α	h	e
$8P_1 + 6P_2 + Q + 4D$	106	$53/2$	0	0	0	-2	8.

For example the T, c, α coefficients are respectively $-8/8 + 1 = 0$, $-8/8 - 6 - 1 + 8 = 0$, and $-8/8 + 1 = 0$; the h and e coefficients are $8(-1/2) + 6(-1/2) + 5 = -2$ and $8(1/2) + 6(1/2) + 1 = 8$. Every one of the four bounds contains the constant -2 , and the normalized weights sum to one. Since $N = 4r + L$ and $e = O - N$, we obtain

$$U \leq \frac{106r + (53/2)L + 8e - 2h}{19} - 2 = \frac{53}{38}N + \frac{8}{19}(O - N) - \frac{2}{19}h - 2.$$

This is (3). Each row bounds one separately computed matching-and-tour algorithm; no step replaces the optimum of a sum by a sum of independent optima. Taking the minimum of the four actual tours is exactly what justifies the convex comparison. The signed value of e_X has already been preserved in each comparison and requires no additional sign assumption here. Polynomiality follows from the explicit constructions and is summarized in Subsection 6.4. $\square$

Corollary 5.7 (Ordinary supplied even factor). *If an ordinary connected unweighted graph has a supplied spanning collection of at least two pairwise vertex-disjoint even unit cycles, the same algorithm satisfies*

$$U \leq \frac{37}{38}n + \frac{8}{19}\text{OPT}(G) - \frac{2}{19}h - 2 \leq \frac{53}{38}\text{OPT}(G) - 2.$$

Additional unit edges, including chords and triangles, are permitted. For one supplied cycle it is solved exactly.

Proof. Apply Theorem 3.2 to its ordinary shortest-path metric and an optimum metric Hamilton cycle, whose cost is $\text{OPT}(G) \geq n$. Expand every edge copy of the returned metric multigraph along a shortest path. Connectivity and spanning are preserved. At each formal endpoint, the expanded paths contribute its even multigraph degree, and every internal occurrence on an expanded path contributes degree two; hence all degrees remain positive and even. The cost is unchanged. Finally, $n \leq \text{OPT}(G)$ and $h \geq 0$ give the second displayed inequality. $\square$

The supplied-factor condition is not assumed for every bipartite graph. The remainder of the proof removes it, rather than invoking a nonexistent strict two-factor on an unbalanced bipartition.

6 Private squares, vertex copies, and the bipartite transfer

6.1 A reversible way to cover every core isolate

Lemma 6.1 (Private-square identity and output projection). *Let I be any set of vertices of a connected unweighted graph G with at least two vertices. For each $v \in I$, add three new vertices a_v, b_v, c_v and exactly the edges $va_v, a_vb_v, b_vc_v, c_vv$, all of unit length. The new vertices have no other incident edges. Write G^+ for the augmented graph and $i = |I|$. Then*

$$\text{OPT}(G^+) = \text{OPT}(G) + 4i.$$

Moreover every graphic tour of G^+ can be converted in polynomial time to a graphic tour of G by deleting all private-square edges, saving at least $4i$.

Proof. A tour of G can traverse the four-edge cycle of each added square once as an excursion from its attachment vertex. The excursions preserve connectivity and even degree, proving $\text{OPT}(G^+) \leq \text{OPT}(G) + 4i$.

Conversely let K^+ be any connected spanning even multigraph in G^+ . For $v \in I$, let R_v be the multiset of edge copies of K^+ on the private square attached at v . Each private vertex has no edge outside that square, while K^+ is spanning and connected. Consequently the support of R_v is connected on $\{v, a_v, b_v, c_v\}$. The degrees of a_v, b_v, c_v in R_v equal their degrees in K^+ and are positive and even. The handshaking identity then makes the degree contributed at v even; connectivity makes it positive. Thus all four degrees in R_v are at least two, and

$$|R_v| = \frac{1}{2} \sum_{x \in \{v, a_v, b_v, c_v\}} \deg_{R_v}(x) \geq 4.$$

Delete every R_v . This subtracts an even degree at each attachment and does not change the degree of any other original vertex. The remaining support connects all original vertices: any path in K^+ that enters a private square can leave it only through the same attachment, so each such excursion can be erased. Because G has at least two vertices, a connected spanning support on its original vertices gives every one of them positive remaining degree. We therefore obtain a graphic tour of G , save at least $4i$ edge copies, and prove $\text{OPT}(G) \leq \text{OPT}(G^+) - 4i$. Together with the first inequality this proves the identity. An output initially given in the shortest-path metric of G^+ is first expanded edge by edge; its cost is unchanged, after which the same projection applies. $\square$

6.2 Copying a whole even-decomposable core

Lemma 6.2 (Whole-core copying). *Let J be a connected simple unweighted graph. Suppose $F^+ \subseteq E(J)$ has positive even degrees at all vertices and is supplied with an edge-disjoint decomposition into simple even cycles, with $f^+ = |F^+| \leq \text{OPT}(J)$. If the decomposition has at least two cycles, one can construct a graphic tour of J of cost at most*

$$\frac{37}{38}f^+ + \frac{8}{19}\text{OPT}(J) - 2. \quad (20)$$

A single spanning even cycle can instead be used directly.

Proof. Make one formal copy of a vertex for each of its occurrences in a cycle of the supplied decomposition, and let π map every copy to its original vertex. The copied cycles are pairwise vertex-disjoint formal bags. Their total number of vertices is

$$N = \sum_{C \text{ in decomposition}} |V(C)| = \sum_C |E(C)| = f^+.$$

Give the copies the pullback pseudometric $d(x, y) = d_J(\pi(x), \pi(y))$. It is integer-valued. Because every decomposition cycle is simple, π is injective within each bag, so distinct vertices of a bag have distance at least one, and its designated cycle edges have distance one. Every original vertex is represented because F^+ has positive degree everywhere. The graph on copies at distance at most one is connected: all copies of one original vertex are mutually joined at distance zero, and any copies of adjacent vertices of J are joined at distance one. Hence the copied data form an admissible formal instance.

Take an optimum Hamilton cycle in the shortest-path metric of J , whose cost is $\text{OPT}(J)$. At the occurrence of each original vertex v , replace v by a consecutive list of all copies in $\pi^{-1}(v)$. Consecutive copies in that list have distance zero, while the entering and leaving edge lengths are unchanged. This gives a Hamilton cycle on the N formal vertices of cost

$$O = \text{OPT}(J) \geq f^+ = N,$$

where the inequality is the hypothesis of the lemma. The formal instance has at least two bags, so Theorem 3.2 applies. Dropping its nonpositive h term gives a formal tour of cost at most

$$\frac{37}{38}N + \frac{8}{19}O - 2 = \frac{37}{38}f^+ + \frac{8}{19}\text{OPT}(J) - 2.$$

Identify copies with the same image under π , delete resulting zero loops, and expand every remaining metric edge copy along a shortest path in J . Identification preserves connectivity, and deleting loops cannot disconnect distinct identified vertices. For an original vertex v , the sum of

the even formal degrees over $\pi^{-1}(v)$ is even; each deleted loop subtracts two, and every occurrence internal to an expanded path adds two. Thus all projected degrees are even.

They are also positive. The algorithms in Proposition 4.4 give every formal vertex positive degree on same-bag unit edges before the quotient tree is added. Such an edge never becomes a loop because π is injective on its bag. Since every original vertex has a formal copy, each retains a genuine incident edge after identification. Expansion preserves cost and can only add intersections, so the result is a connected spanning even multigraph in J with the displayed bound. This proves (20).

If the supplied decomposition has one cycle, positivity of the degrees of F^+ implies that this simple even cycle spans J . Returning it costs $|V(J)|$, which is optimal by (1). $\square$

The completeness of the representation is important. Copying only part of a component while leaving some original vertices without a represented positive-degree occurrence would not justify this projection argument. Likewise, arbitrary zero-cost cycles are not used as local bags.

6.3 Assembly of the main bound

Proof of Theorem 2.2. For one vertex return the empty tour. If G is a tree, every edge is a bridge. In any connected spanning even multigraph, the number of edge copies crossing either shore of a cut is even, because it has the same parity as the sum of degrees on that shore. A bridge cut must also be crossed positively, so every tree edge is used at least twice. Doubling the tree has cost $2(n-1)$ and is therefore optimal. This includes the two-vertex case.

Now assume G contains a cycle. Compute a minimum- Φ core F using Proposition 2.3, which gives

$$F \neq \emptyset, \quad f + 2i = \Phi \leq \text{OPT}(G), \quad f \leq \text{OPT}(G).$$

Use Lemma 2.4 to decompose F into simple even cycles. For every one of its i isolates attach a private square as in Lemma 6.1. The enlarged graph G^+ is still connected and bipartite: color a_v, c_v opposite to v and b_v the same as v . The edge set F^+ consisting of F and all private squares has positive even degrees at every vertex. Indeed, the nonisolated original vertices retain their positive even F -degrees, while every isolate and every new private vertex has degree two in its square. It has the explicit edge-disjoint even-cycle decomposition consisting of the cycles of F and the i private squares. In particular, its number of cycles is $|\mathcal{C}| + i$, and

$$|F^+| = f + 4i, \quad \text{OPT}(G^+) = \text{OPT}(G) + 4i.$$

The core lower bound $f \leq \text{OPT}(G)$ therefore gives $|F^+| \leq \text{OPT}(G^+)$, exactly the comparison-cost hypothesis of Lemma 6.2. There is no requirement that F^+ minimize the core objective in the augmented graph.

If this decomposition has at least two cycles, apply Lemma 6.2 and then remove every private square from the actual expanded tour. The cost of the resulting original-graph tour is at most

$$\begin{aligned} U &\leq \frac{37}{38}(f + 4i) + \frac{8}{19}(\text{OPT}(G) + 4i) - 4i - 2 \\ &= \frac{37}{38}f + \frac{8}{19}\text{OPT}(G) + \frac{30}{19}i - 2 \\ &= \frac{37}{38}\Phi + \frac{8}{19}\text{OPT}(G) - \frac{7}{19}i - 2. \end{aligned} \tag{21}$$

This is the first inequality in (2). Since $\Phi \leq \text{OPT}(G)$ and $i \geq 0$, it also gives

$$U \leq \left(\frac{37}{38} + \frac{8}{19}\right)\text{OPT}(G) - 2 = \frac{53}{38}\text{OPT}(G) - 2.$$

The retained coefficient of N in the formal theorem is essential: using only a coarse $53/38$ multiplicative bound before deleting the private squares would not give this accounting.

If the enlarged decomposition has just one cycle, then $i = 0$: Proposition 2.3 makes F nonempty, so any isolate would add a private square to at least one cycle already present in F . Thus $\mathcal{C}$ itself consists of one simple cycle. Since $i = 0$, that cycle contains every vertex of G and is a spanning even unit cycle. It is an optimal tour of length n by (1). For even $n \geq 6$ this exact output also satisfies $n \leq (53/38)n - 2$, because $15n/38 \geq 2$. For $n = 4$ that additive inequality is false, so use the stated single-square exception. In every case the multiplicative ratio $53/38$ holds.

All constructed auxiliary graphs and outputs have polynomial size, and all searches were explicitly polynomial. The execution order and bounds are summarized next. No unproved structural lemma about an arbitrary bipartite input is needed: the only use of bipartiteness is that the computed binary even core decomposes into even cycles. $\square$

6.4 Algorithm and running time

Fix deterministic tie-breaking. Handle a singleton or tree directly; otherwise compute the minimum core of Proposition 2.3, decompose it, add the private squares, and return the cycle if the augmented decomposition has only one. For at least two cycles, form the copied formal instance, compute shortest paths and the certificates of Subsection 3.2, and construct P_1, P_2, Q, D . Each of the first three solves one global matching, decodes its crossings, performs the switches, and doubles a quotient-tree; D is direct. Project the cheapest tour by identifying copies, expanding stored paths, and deleting the private squares.

The relevant size bounds are

object	vertices	edges
core matching graph	$4m$	$O(m^2)$
G^+	$n + 3i \leq 4n$	$m + 4i \leq m + 4n$
formal instance	$N = f + 4i \leq m + 4n$	$\binom{N}{2}$
each global matching graph	$O(N)$	$O(N^2)$.

After scaling by four, all matching weights are polynomial-bit integers. Breadth-first search supplies the metric paths; the width-five certificate DP has a constant number of states per bag, and the $k = 4, 6$ checks have fixed size. Every switch strictly reduces a component count at most N , so fewer than N occur. Path expansion and square deletion are linear in their polynomial-size outputs. Thus the algorithm is deterministic and polynomial. The comparator H , its parameters, and the half-marginal distribution are used only in the analysis, never as algorithmic inputs.

Use of AI: We prepared a list of proof strategies and techniques related to approximating TSP. We fed this to multiple AI tools (Claude, ChatGPT and two open-weight models with open-source harnesses) and prompted these AI tools to improve the existing approximation bounds. When an AI tool was hallucinating, stuck, or going in the wrong direction, we stopped that AI tool, corrected the intermediate .md files, added more specific new ideas and switched to a different AI tool with more detailed fresh prompts. Equations written by one AI tool were critiqued, verified and expanded by a completely different AI tool in an iterative manner, till we were satisfied with their correctness and details. The author is solely responsible for the correctness of the proofs.

References

- [1] B. Candráková and R. Lukotka. *Cubic TSP—A 1.3-Approximation*. SIAM Journal on Discrete Mathematics, 32(3):2094–2114, 2018. <https://doi.org/10.1137/15M1032247>.
- [2] N. Christofides. *Worst-Case Analysis of a New Heuristic for the Travelling Salesman Problem*. Management Sciences Research Report 388, Carnegie-Mellon University, 1976.
- [3] A. Çivril. *4/3-Approximation of Graphic TSP*. Withdrawn preprint, arXiv:2305.05411; withdrawn November 4, 2024, with the latest withdrawal note dated July 21, 2026. <https://arxiv.org/abs/2305.05411>.
- [4] Z. Dvořák, D. Král', and B. Mohar. *Graphic TSP in Cubic Graphs*. Proceedings of STACS 2017, LIPIcs 66, Article 27, 2017. <https://doi.org/10.4230/LIPIcs.STACS.2017.27>.
- [5] J. Edmonds. *Maximum Matching and a Polyhedron with 0,1-Vertices*. Journal of Research of the National Bureau of Standards B, 69B:125–130, 1965. https://nvlpubs.nist.gov/nistpubs/jres/69B/jresv69Bn1-2p125_A1b.pdf.
- [6] A. R. Karlin, N. Klein, and S. Oveis Gharan. *A (Slightly) Improved Approximation Algorithm for Metric TSP*. Proceedings of the 53rd ACM Symposium on Theory of Computing, pages 32–45, 2021. <https://arxiv.org/abs/2007.01409>.
- [7] J. A. Karp and R. Ravi. *A 9/7-Approximation Algorithm for Graphic TSP in Cubic Bipartite Graphs*. Discrete Applied Mathematics, 209:164–216, 2016. <https://arxiv.org/abs/1311.3640>.
- [8] T. Mömke and O. Svensson. *Removing and Adding Edges for the Traveling Salesman Problem*. Journal of the ACM, 63(1), Article 2, 2016. <https://doi.org/10.1145/2739008>.
- [9] M. Mucha. *13/9-Approximation for Graphic TSP*. Theory of Computing Systems, 55(4):640–657, 2014. <https://arxiv.org/abs/1108.1130>.
- [10] A. Newman. *An Improved Analysis of the Mömke–Svensson Algorithm for Graph-TSP on Subquartic Graphs*. SIAM Journal on Discrete Mathematics, 34(1):865–884, 2020. <https://arxiv.org/abs/1407.2524>.
- [11] S. Oveis Gharan, A. Saberi, and M. Singh. *A Randomized Rounding Approach to the Traveling Salesman Problem*. Proceedings of the 52nd IEEE Symposium on Foundations of Computer Science, pages 550–559, 2011. <https://doi.org/10.1109/FOCS.2011.80>.
- [12] A. Sebő and J. Vygen. *Shorter Tours by Nicer Ears: 7/5-approximation for graphic TSP, 3/2 for the path version, and 4/3 for two-edge-connected subgraphs*. Combinatorica, 34(5):597–629, 2014. <https://arxiv.org/abs/1201.1870>.
- [13] A. I. Serdyukov. *On Some Extremal Walks in Graphs* (in Russian). Upravlyaemye Sistemy, 17:76–79, 1978. <https://www.mathnet.ru/eng/da1078>.
- [14] A. van Zuylen. *Improved Approximations for Cubic and Cubic Bipartite TSP*. Mathematical Programming, 172(1–2):399–413, 2018. <https://arxiv.org/abs/1507.07121>.
- [15] M. C. Wigal, Y. Yoo, and X. Yu. *Approximating TSP Walks in Subcubic Graphs*. Journal of Combinatorial Theory, Series B, 158:70–104, 2023. <https://doi.org/10.1016/j.jctb.2022.09.002>.